

Grade 11 - Term 3

C1–C4 Practice Exercises and Solutions

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Evaluated criteria

- C1** Characterises continuous random variables by distinguishing them from discrete variables and interpreting their probability density function, cumulative distribution function, and support in economic contexts.
- C2** Applies uniform and triangular continuous distributions with constant or piecewise-linear probability density functions to calculate interval probabilities using geometric areas.
- C3** Applies circular-arc probability density functions to calculate interval probabilities using the geometry of circles and circular segments.
- C4** Applies the exponential distribution to constant-rate waiting-time situations and assesses its suitability.

1 Distribution Recognition in Small Businesses [C1-8]

Consider the following four situations faced by small businesses and financial organisations.

1. A small-business lender wants to study the risk associated with its loan portfolio. As part of its monitoring process, the lender randomly selects one loan applicant and records whether that applicant defaults within the following year. For the selected applicant, the recorded outcome is either default or no default.
2. A sales representative contacts potential customers by telephone as part of a regular sales campaign. The representative continues making calls until the first successful sale is obtained and records the total number of calls required. The calls are treated as independent, and the probability of making a sale is assumed to remain the same from one call to the next.

3. A small retailer depends on a supplier for regular product deliveries. When an order is placed, the supplier estimates that the delivery may arrive at any time between 2 and 5 days after the order, with no particular time within that interval considered more likely than another.
4. A workshop produces customised products for local businesses. Based on previous experience, a particular type of job takes between 30 and 50 minutes to complete. Completion times near 40 minutes occur most frequently, while times become progressively less common as they approach either the minimum or maximum.

Questions/Tasks.

- | | |
|---|-----|
| C1 Classify the loan-default outcome. | [2] |
| C1 Classify the number of sales calls. | [2] |
| C1 Classify the supplier's delivery time. | [2] |
| C1 Classify the workshop completion time. | [2] |

C1 - Classify the loan-default outcome. [2]

The random variable is **discrete** and follows a **Bernoulli distribution**. It represents one trial with exactly two possible outcomes: default or no default.



C1 - Classify the number of sales calls. [2]

The random variable is **discrete** and follows a **geometric distribution**. It counts the number of independent trials needed to obtain the first success when the success probability remains the same.



C1 - Classify the supplier's delivery time. [2]

The random variable is **continuous** and follows a **uniform distribution**. Its support is the bounded interval from 2 to 5 days, and every value in that interval is equally plausible.



C1 - Classify the workshop completion time. [2]

The random variable is **continuous** and follows a **triangular distribution**. It has a stated minimum, maximum, and most likely value, with likelihood changing steadily toward and away from the mode.



2 Distribution Recognition in Economic Operations [C1-8]

Consider the following four situations arising in routine economic and business operations.

1. An auditing firm is reviewing the accuracy of a company's invoicing process. During each review, an auditor selects exactly 20 invoices and records how many of them contain an error. Each invoice is inspected separately, and the chance of finding an error is assumed to be the same for each invoice.
2. A retail shop wants to understand customer traffic so that it can plan staffing levels more effectively. The shop records the number of customers who arrive during each fixed 10-minute interval. During the period being studied, arrivals occur separately from one another and the average arrival rate remains approximately stable.
3. A company is analysing how long employees take to process invoices. The processing time T can take any value from 0 to 10 minutes. The model indicates that larger processing times within this interval occur with progressively greater density.
4. A retailer studies the amount spent by individual customers during a typical day. Historical transaction data show that spending values tend to cluster around an average of \$45. Spending amounts above and below this value occur in an approximately symmetric pattern, while amounts far from the average become progressively less common.

Questions/Tasks.

- | | |
|---|-----|
| C1 Classify the number of erroneous invoices. | [2] |
| C1 Classify the number of customer arrivals. | [2] |
| C1 Classify the invoice-processing time. | [2] |
| C1 Classify daily customer spending. | [2] |

C1 - Classify the number of erroneous invoices. [2]

The random variable is **discrete** and follows a **binomial distribution**. It counts successes across a fixed number of independent binary trials with the same success probability.



C1 - Classify the number of customer arrivals. [2]

The random variable is **discrete** and follows a **Poisson distribution**. It counts independent events during a fixed interval when their average rate is approximately constant.



C1 - Classify the invoice-processing time. [2]

The random variable is **continuous** and follows a **linear distribution**. Its pdf changes linearly with time over the bounded support $0 \leq T \leq 10$.



C1 - Classify daily customer spending. [2]

The random variable is **continuous** and follows a **normal distribution**. The distribution is approximately symmetric, is concentrated around its mean, and has a bell-shaped form.



3 Uniform Delivery Time

[C2-9]

A local delivery company records the time required for a particular short-distance route. Under normal traffic conditions, the delivery time ranges from 20 to 50 minutes, and every time within this interval is considered equally plausible. Let T represent the delivery time in minutes.

Questions/Tasks.

C2 Determine the density constant and write the pdf. [3]

C2 Find the probability that a delivery takes between 25 and 40 minutes. [3]

C2 Find the probability that a delivery takes more than 35 minutes. [3]

C2 - Determine the density constant and write the pdf. [3]

Since all values from 20 to 50 minutes are equally plausible, the density is constant on this interval. Let the height of the density be k .

The region under the density is a rectangle with width

$$B = 50 - 20 = 30$$

and height

$$h = k.$$

Since the total area under a probability density function must be 1,

$$A_{\text{rectangle}} = Bh = 1.$$

Therefore,

$$30k = 1 \quad \longrightarrow \quad k = \frac{1}{30}.$$

Hence the pdf is

$$f(t) = \begin{cases} \frac{1}{30}, & 20 \leq t \leq 50, \\ 0, & \text{otherwise.} \end{cases}$$



C2 - Find the probability that a delivery takes between 25 and 40 minutes. [3]

We need to find

$$P(25 \leq T \leq 40).$$

The corresponding region under the density is a rectangle with width

$$B = 40 - 25 = 15$$

and height

$$h = \frac{1}{30}.$$

Therefore,

$$\begin{aligned} P(25 \leq T \leq 40) &= A_{\text{rectangle}} \\ &= Bh \\ &= (15) \left(\frac{1}{30} \right) \\ &= \frac{1}{2}. \end{aligned}$$

Thus,

$$P(25 \leq T \leq 40) = \frac{1}{2}.$$



C2 - Find the probability that a delivery takes more than 35 minutes.

[3]

We need to find

$$P(T > 35).$$

The region from 35 to 50 is a rectangle with width

$$B = 50 - 35 = 15$$

and height

$$h = \frac{1}{30}.$$

Therefore,

$$\begin{aligned} P(T > 35) &= A_{\text{rectangle}} \\ &= (15) \left(\frac{1}{30} \right) \\ &= \frac{1}{2}. \end{aligned}$$

Thus,

$$P(T > 35) = \frac{1}{2}.$$



4 Triangular Production Time

[C2-9]

A workshop models the time required to complete a particular production task. The task cannot take less than 10 minutes or more than 40 minutes. A completion time of 20 minutes is the most likely value. Times become gradually more likely from 10 to 20 minutes and gradually less likely from 20 to 40 minutes. Let T represent the completion time in minutes.

Questions/Tasks.

C2 Determine the height of the triangular density and write the pdf. [3]

C2 Find the probability that the task takes less than 20 minutes. [3]

C2 Find the probability that the task takes between 20 and 30 minutes. [3]

C2 - Determine the height of the triangular density and write the pdf. [3]

The support extends from 10 to 40, and the peak occurs at 20. Let the height of the peak be h .

The complete region under the density is a triangle with base

$$B = 40 - 10 = 30.$$

Since its area must equal 1,

$$\frac{1}{2}(30)h = 1.$$

Therefore,

$$15h = 1 \quad \longrightarrow \quad h = \frac{1}{15}.$$

Left-hand piece.

From $(10, 0)$ to $(20, \frac{1}{15})$, the slope is

$$m = \frac{\frac{1}{15} - 0}{20 - 10} = \frac{1}{150}.$$

Therefore,

$$f(t) = \frac{t - 10}{150} \quad \text{for } 10 \leq t \leq 20.$$

Right-hand piece.

From $(20, \frac{1}{15})$ to $(40, 0)$, the slope is

$$m = \frac{0 - \frac{1}{15}}{40 - 20} = -\frac{1}{300}.$$

Therefore,

$$f(t) = \frac{40 - t}{300} \quad \text{for } 20 \leq t \leq 40.$$

Hence,

$$f(t) = \begin{cases} \frac{t - 10}{150}, & 10 \leq t \leq 20, \\ \frac{40 - t}{300}, & 20 \leq t \leq 40, \\ 0, & \text{otherwise.} \end{cases}$$



C2 - Find the probability that the task takes less than 20 minutes. [3]

The region from 10 to 20 is a triangle with base

$$B = 20 - 10 = 10$$

and height

$$h = \frac{1}{15}.$$

Therefore,

$$\begin{aligned} P(T < 20) &= A_{\triangle} \\ &= \frac{1}{2}(10) \left(\frac{1}{15} \right) \\ &= \frac{1}{3}. \end{aligned}$$

Thus,

$$P(T < 20) = \frac{1}{3}.$$



C2 - Find the probability that the task takes between 20 and 30 minutes.

[3]

We need to find

$$P(20 \leq T \leq 30).$$

At $t = 20$,

$$f(20) = \frac{1}{15}.$$

At $t = 30$,

$$f(30) = \frac{40 - 30}{300} = \frac{1}{30}.$$

The region from 20 to 30 forms a trapezoid with width

$$B = 30 - 20 = 10$$

and heights

$$h_1 = \frac{1}{15}, \quad h_2 = \frac{1}{30}.$$

Therefore,

$$\begin{aligned} P(20 \leq T \leq 30) &= A_{\text{trapezoid}} \\ &= \frac{1}{2}(10) \left(\frac{1}{15} + \frac{1}{30} \right) \\ &= 5 \left(\frac{3}{30} \right) \\ &= \frac{1}{2}. \end{aligned}$$

Thus,

$$P(20 \leq T \leq 30) = \frac{1}{2}.$$



5 Semicircular Density Model

[C3-9]

A continuous random variable X is modeled using the upper half of a circle. The density is symmetric around $x = 0$ and has support from -2 to 2 . Its graph is a scaled semicircle.

Questions/Tasks.

C3 Determine the scale factor required for the semicircle to define a probability density function. [3]

C3 Find $P(-2 \leq X \leq 0)$. [3]

C3 Find $P(0 \leq X \leq 1)$ using circular-segment geometry. [3]

C3 - Determine the scale factor required for the semicircle to define a probability density function. [3]

The unscaled curve is the upper semicircle

$$y = \sqrt{4 - x^2},$$

which has radius

$$r = 2.$$

The area of a complete circle is

$$A_{\text{circle}} = \pi r^2 = 4\pi.$$

Therefore, the area of the upper semicircle is

$$A_{\text{semicircle}} = \frac{1}{2}(4\pi) = 2\pi.$$

Suppose the density is

$$f(x) = k\sqrt{4 - x^2}.$$

The total area must equal 1, so

$$k(2\pi) = 1.$$

Therefore,

$$k = \frac{1}{2\pi}.$$

Hence,

$$f(x) = \begin{cases} \frac{1}{2\pi}\sqrt{4 - x^2}, & -2 \leq x \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$



C3 - Find $P(-2 \leq X \leq 0)$. [3]

The density is symmetric around $x = 0$. Therefore, the region from -2 to 0 is exactly half of the total density region.

Since the total area is 1,

$$P(-2 \leq X \leq 0) = \frac{1}{2}.$$



C3 - Find $P(0 \leq X \leq 1)$ using circular-segment geometry.

[3]

Consider the unscaled circle

$$x^2 + y^2 = 4.$$

At $x = 1$,

$$1^2 + y^2 = 4 \quad \longrightarrow \quad y = \sqrt{3}.$$

The point

$$(1, \sqrt{3})$$

lies on the circle of radius 2.

The corresponding central angle from the positive x -axis is

$$\theta = \frac{\pi}{3}.$$

The region under the arc from $x = 0$ to $x = 1$ can be decomposed into a sector and a right triangle.

The sector angle is

$$\frac{\pi}{2} - \frac{\pi}{3} = \frac{\pi}{6}.$$

Therefore, the sector area is

$$A_{\text{sector}} = \frac{1}{2}r^2\theta = \frac{1}{2}(2^2)\left(\frac{\pi}{6}\right) = \frac{\pi}{3}.$$

The right triangle has base 1 and height $\sqrt{3}$, so

$$A_{\triangle} = \frac{1}{2}(1)(\sqrt{3}) = \frac{\sqrt{3}}{2}.$$

Thus, the unscaled area under the arc is

$$A = \frac{\pi}{3} + \frac{\sqrt{3}}{2}.$$

Since the density is scaled by

$$\frac{1}{2\pi},$$

the probability is

$$P(0 \leq X \leq 1) = \frac{1}{2\pi} \left(\frac{\pi}{3} + \frac{\sqrt{3}}{2} \right).$$

Therefore,

$$\boxed{P(0 \leq X \leq 1) = \frac{1}{6} + \frac{\sqrt{3}}{4\pi}}$$



6 Circular-Arc Measurement Model

[C3-9]

A continuous measurement X is modeled by a symmetric circular-arc density supported from -3 to 3 . Before normalization, the graph follows the upper half of the circle

$$x^2 + y^2 = 9.$$

Questions/Tasks.

C3 Determine the normalized pdf.

[3]

C3 Find $P(X > 0)$.

[3]

C3 Find $P(\frac{3}{2} \leq X \leq 3)$ using a circular segment.

[3]

C3 - Determine the normalized pdf.

[3]

The circle has radius

$$r = 3.$$

The area of the upper semicircle is

$$A_{\text{semicircle}} = \frac{1}{2}\pi r^2 = \frac{9\pi}{2}.$$

Let

$$f(x) = k\sqrt{9 - x^2}.$$

Since the total area must equal 1,

$$k \left(\frac{9\pi}{2} \right) = 1.$$

Therefore,

$$k = \frac{2}{9\pi}.$$

Hence,

$$f(x) = \begin{cases} \frac{2}{9\pi}\sqrt{9 - x^2}, & -3 \leq x \leq 3, \\ 0, & \text{otherwise.} \end{cases}$$



C3 - Find $P(X > 0)$.

[3]

The density is symmetric around $x = 0$. Therefore,

$$P(X > 0) = \frac{1}{2}.$$



C3 - Find $P(\frac{3}{2} \leq X \leq 3)$ using a circular segment.

[3]

At

$$x = \frac{3}{2},$$

the corresponding point on the circle satisfies

$$\left(\frac{3}{2}\right)^2 + y^2 = 9.$$

Therefore,

$$y^2 = 9 - \frac{9}{4} = \frac{27}{4},$$

so

$$y = \frac{3\sqrt{3}}{2}.$$

The point

$$\left(\frac{3}{2}, \frac{3\sqrt{3}}{2}\right)$$

corresponds to the central angle

$$\theta = \frac{\pi}{3}.$$

The region under the arc from $\frac{3}{2}$ to 3 is a circular segment.

The sector area is

$$A_{\text{sector}} = \frac{1}{2}r^2\theta = \frac{1}{2}(9)\left(\frac{\pi}{3}\right) = \frac{3\pi}{2}.$$

The triangle formed by the two radii has area

$$A_{\triangle} = \frac{1}{2}r^2 \sin \theta = \frac{1}{2}(9) \sin\left(\frac{\pi}{3}\right).$$

Therefore,

$$A_{\triangle} = \frac{9}{2} \left(\frac{\sqrt{3}}{2}\right) = \frac{9\sqrt{3}}{4}.$$

Thus, the circular-segment area is

$$A_{\text{segment}} = \frac{3\pi}{2} - \frac{9\sqrt{3}}{4}.$$

Applying the normalization factor,

$$P\left(\frac{3}{2} \leq X \leq 3\right) = \frac{2}{9\pi} \left(\frac{3\pi}{2} - \frac{9\sqrt{3}}{4}\right).$$

Therefore,

$$\boxed{P\left(\frac{3}{2} \leq X \leq 3\right) = \frac{1}{3} - \frac{\sqrt{3}}{2\pi}}$$



7 Customer Arrival Waiting Time

[C4-9]

During a quiet period at a service desk, customers arrive independently at an approximately constant average rate of 6 customers per hour. Let T represent the waiting time, in minutes, until the next customer arrives.

Questions/Tasks.

C4 Determine the exponential rate parameter in minutes and write the pdf. [3]

C4 Find the probability that the next customer arrives within 5 minutes. [3]

C4 Explain whether an exponential model is suitable in this situation. [3]

C4 - Determine the exponential rate parameter in minutes and write the pdf. [3]

The average arrival rate is

6 customers per hour.

Since one hour contains 60 minutes,

$$\lambda = \frac{6}{60} = \frac{1}{10}$$

customer per minute.

Therefore,

$$\lambda = 0.1.$$

For an exponential waiting-time distribution,

$$f(t) = \lambda e^{-\lambda t}.$$

Hence,

$$f(t) = \begin{cases} 0.1e^{-0.1t}, & t \geq 0, \\ 0, & t < 0. \end{cases}$$



C4 - Find the probability that the next customer arrives within 5 minutes. [3]

For an exponential random variable,

$$P(T \leq t) = 1 - e^{-\lambda t}.$$

Therefore,

$$\begin{aligned} P(T \leq 5) &= 1 - e^{-(0.1)(5)} \\ &= 1 - e^{-0.5}. \end{aligned}$$

Thus,

$$P(T \leq 5) = 1 - e^{-0.5}$$

and approximately,

$$P(T \leq 5) \approx 0.3935.$$



C4 - Explain whether an exponential model is suitable in this situation.

[3]

An exponential model is suitable when arrivals can reasonably be treated as independent and when the average arrival rate remains approximately constant during the period being modeled.

The problem states that customers arrive independently and that the average rate is approximately constant at 6 customers per hour during the quiet period. These conditions support the use of an exponential waiting-time model.

Therefore, the exponential distribution is a reasonable model for this situation, provided that the arrival rate remains approximately constant during the period considered.



8 Technical Support Waiting Time

[C4-9]

A technical-support center receives calls throughout the day. Historical records show an overall average of 12 calls per hour. However, the center is much busier during the morning and immediately after lunch than during the late afternoon. A manager proposes using a single exponential distribution with rate 12 calls per hour to model the waiting time until the next call at any time during the working day.

Questions/Tasks.

C4 Using the proposed model, determine the rate parameter in minutes and the expected waiting time. [3]

C4 Using the proposed model, find the probability that the waiting time exceeds 10 minutes. [3]

C4 Assess whether a single exponential distribution is suitable for modeling waiting times throughout the entire day. [3]

C4 - Using the proposed model, determine the rate parameter in minutes and the expected waiting time. [3]

The proposed average rate is

12 calls per hour.

Converting this to calls per minute,

$$\lambda = \frac{12}{60} = \frac{1}{5}.$$

Therefore,

$$\lambda = 0.2$$

calls per minute.

For an exponential distribution,

$$E(T) = \frac{1}{\lambda}.$$

Therefore,

$$E(T) = \frac{1}{0.2} = 5.$$

Thus, according to the proposed model, the expected waiting time is

5 minutes.



C4 - Using the proposed model, find the probability that the waiting time exceeds 10 minutes. [3]

For an exponential random variable,

$$P(T > t) = e^{-\lambda t}.$$

Therefore,

$$\begin{aligned} P(T > 10) &= e^{-(0.2)(10)} \\ &= e^{-2}. \end{aligned}$$

Thus,

$$P(T > 10) = e^{-2}$$

and approximately,

$$P(T > 10) \approx 0.1353.$$



C4 - Assess whether a single exponential distribution is suitable for modeling waiting times throughout the entire day. [3]

A single exponential waiting-time model assumes that the underlying event rate is constant during the period being modeled.

In this situation, the call rate changes substantially throughout the day. The center is busier during the morning and immediately after lunch and quieter during the late afternoon. Therefore, the constant-rate assumption required by a single exponential model is not well supported.

The overall average of 12 calls per hour can describe the average number of calls across the day, but it does not imply that calls arrive at that same rate during every part of the day.

Therefore, a single exponential distribution with rate 12 calls per hour is not an appropriate model for waiting times throughout the entire working day. Separate models for periods with approximately constant call rates would be more appropriate.

